对单个质点在保守场中运动.

势能 $U = U(x)$

势能对一阶导数为0的点都是平衡点.

$$U'(x_0) = 0$$

其中,

极小值 ($U''(x_0) < 0$):

不稳定的平衡点, 微小扰动质点便会远离)

极大值 ($U''(x_0) > 0$):

稳定的平衡点.

在其附近质点可围绕其作小振动.

求振动的周期(频率)

设 $x = x_0$ 处为稳定平衡点

$$U'(x_0) = 0 \quad ; \quad U''(x_0) > 0$$

对 $U(x_0)$ 做泰勒展开:

$$U(x) = U(x_0) + \underbrace{U'(x_0)}_{=0}(x-x_0) + \frac{1}{2}U''(x_0)(x-x_0)^2 + \cdots$$

$$= U(x_0) + \frac{1}{2}U''(x_0)(x-x_0)^2 + \cdots$$

对小振动, 取近似忽略 $(x-x_0)^3$ 以上各项.

$$U(x) = U(x_0) + \frac{1}{2}U''(x_0)(x-x_0)^2$$

选取势能参考点，令 $U(x_0)=0$，$x_0=0$，于是有：

$$U(x)=\frac{1}{2}U''(x_0)(x-x_0)^2$$

机械能守恒：

$$\frac{1}{2}mv^2=E-\frac{1}{2}U''(x_0)x^2=E-\frac{1}{2}U_0''x^2$$

$$\Longleftrightarrow \quad m\dot{x}^2=2E-U_0''x^2$$

分离变量：

$$\frac{dx}{dt}=\sqrt{\frac{2E-U_0''x^2}{m}}$$

$$\frac{dx}{\sqrt{1-\frac{U_0''}{2E}x^2}}=\sqrt{\frac{2E}{m}}\,dt$$

换元：

$$\sqrt{\frac{U_0''}{2E}}x=\sin\varphi,\qquad dx=\sqrt{\frac{2E}{U_0''}}\cos\varphi\,d\varphi,\qquad \sqrt{1-\frac{U_0''}{2E}x^2}=\cos\varphi$$

$$\frac{dx}{\sqrt{1-\frac{U_0''}{2E}x^2}}=\sqrt{\frac{2E}{U_0''}}\,d\varphi=\sqrt{\frac{2E}{m}}\,dt$$

$$\Longleftrightarrow \quad \varphi=\sqrt{\frac{U_0''}{m}}t+\varphi_0$$

$$x=\sqrt{\frac{2E}{U_0''}}\sin\varphi=\sqrt{\frac{2E}{U_0''}}\sin\left(\sqrt{\frac{U_0''}{m}}t+\varphi_0\right)$$

周期：$T=\dfrac{2\pi}{\sqrt{\frac{U_0''}{m}}}=2\pi\sqrt{\dfrac{m}{U_0''}}$